RW–962

Sub. Code
541101/ 545101

M.C.A. (R)/WE DEGREE EXAMINATION, NOVEMBER 2017

First Semester

DIGITAL COMPUTER ORGANIZATION

(Common for M.C.A. (R) /M.C.A. (WE))

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A $(10 \times 2 = 20)$

Answer **all** the questions.

1. What are Micro-computers?

2. Convert the binary number $(1010.011)_2$ to decimal.

3. Define: Encoder.

4. What are Registers?

5. Write a note on: MAR.

6. List any two common operations of execute cycle.

7. What is memory stack?

8. What is Indexed Addressing?

9. Comment on Interface Chip.

10. What is meant by cycle stealing?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Explain the fundamental concepts of Boolean Algebra.

Or

(b) Write short note on Two level implementation of combinational circuits.

12. (a) Write short notes on multiplexer.

Or

(b) Explain fixed point representation with examples.

13. (a) Write short note on instruction codes.

Or

(b) Explain the design of basic computer.

14. (a) Write short note on Instruction Formats.

Or

(b) Explain the Instructions for Data Transfer and Manipulation.

15. (a) Write short note on Input Output interface.

Or

(b) Explain Memory Management Hardware.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the concept of sum of products and products of sum with examples.

17. Elaborate Error detection codes with an example.

18. Explain the design of Accumulator logic.

19. Elaborate the concept of Stack Organization.

20. Explain in detail about serial communication.

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RW–963

Sub. Code
541102/ 545102

M.C.A. (R)/WE DEGREE EXAMINATION, NOVEMBER 2017.

First Semester

Computer Application

C AND DATA STRUCTURES

(CBCS – 2017 onwards)

(Common for M.C.A. (R)/M.C.A. (WE))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write a note on : C Data types.
2. What are arithmetic operators in C?
3. Define : Array.
4. What are pointers?
5. What is stack?
6. Comment on polish notation.
7. Define : Binary trees.
8. What are non-linear data structures?
9. What is searching?
10. What is Bubble sort?

Part B

$(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the rules for defining variables in C? Give examples.

Or

(b) Write about scanf ( ) and print ( ) functions with examples.

12. (a) What are functions? Discuss about different categories of functions.

Or

(b) Write about pointers and arrays in C with example.

13. (a) How to insert an element in a singly linked list? Give example.

Or

(b) Discuss about queue and its operations.

14. (a) Write short notes on : Types of Binary trees.

Or

(b) Explain about traversing binary trees.

15. (a) What is linear search? Give an example.

Or

(b) Write short notes on selection sort.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Explain about various control structures in C with suitable examples.

17. Write a C program to perform addition and subtraction of two input matrices.

18. Explain about various operations on a stack with examples.

19. Describe any two hashing techniques with examples.

20. Explain the working of insertion sort with an example.

RW–964

Sub. Code
541103/ 545103

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Computer Applications

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(Common for M.C.A. (R), M.C.A. (WE))

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is DDL?
2. Comment on ER Diagram.
3. What is meant by Integrity constraint?
4. Write a note on selection operation in Relational Algebra.
5. What are Nested Queries?
6. Define : Normal Form.
7. What is Transaction? List down the ACID properties.
8. Distinguish between physical and logical undo.
9. What are primary indices?
10. What is a Hash function?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the differences in between Database system and File system.

Or

(b) Write down the additional features of ER Model.

12. (a) Write a note on Enforcing Integrity constraints.

Or

(b) Explain : Tuple Relational Calculus with examples.

13. (a) What are comparison operators in SQL? Explain with examples.

Or

(b) Explain Fourth Normal form with an example.

14. (a) Write about Implementation of Atomicity and Durability.

Or

(b) Write short notes on Buffer Management.

15. (a) Write a note on File organizations.

Or

(b) Write about the structure of B^+–tree with an example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Write about :

    (a) views of data (5)

    (b) instances and schemas. (5)

17. Explain the following Relational Algebra operations with examples.

    (a) set operations (5)

    (b) joins. (5)

18. Describe in detail, about Lossless Join Decomposition.

19. Explain Time stamp-based protocols in detail.

20. Discuss, Indexed Sequential Access methods in detail.

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RW–965

Sub. Code
541104/ 545104

M.C.A. (R/WE) DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Computer Applications

DISCRETE MATHEMATICS

(Common for M.C.A. (R), M.C.A. (WE))

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** the questions.

1. Construct the truth table for $(P \vee Q) \vee \rceil P$.

2. Symbolize the expression "All the world loves a lover".

3. Define an antisymmetric relation with an example.

4. Let $X = \{1,2,3\}, y = \{p,q\}$ and $Z = \{a,b\}$ also let $f : x \to y$ be $f = \{<1,p>,<2,p>,<3,q>\}$ and $g : y \to z$ be given by $g = \{<p,b>,<q,b>\}$ find $g \circ f$.

5. Define a semigroup with an example.

6. If $(G,*)$ is an abelian group, then for all $a,h \in G$ show that $(a \times b)^2 = a^2 * b^2$.

7. Draw a graph with five vertices and seven edges.

8. Define a strongly connected graph with an example.

9. Find the probability of drawing an ace from a standard deck of cards.

10. The sample space $S = \{a,b,c\}$, Assume that $p(a) = 0.2$ and $p(b) = 0.3$ what is $p(c)$?

Section B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \vee R) \vee (P \wedge R) \Leftrightarrow R$.

Or

(b) Show that $(\exists x)\, m(x)$ follows logically from the premises

$(x)(H(x)) \rightarrow m(x)$ and $(\exists x)\, H(x)$.

12. (a) Let $P = \{(1,2),(2,4),(3,3)\}$ and $Q = \{(1,3), (2,4),(4,2)\}$. Find $P \cup Q, P \cap Q, D(P), P(Q)$.

Or

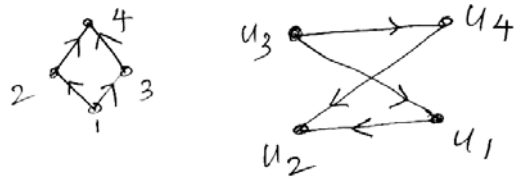
(b) Using characteristics functions prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

13. (a) Show that there exists a homomorphism from $(N,+)$ to $(Z_4,+_4)$ where N is the set of natural numbers and + is the operation of addition on N.

Or

(b) Prove that a subet $S \neq \phi$ of G is a subgroup of $(G,*)$ it and only if for any pair of elements $a,bt\,S$, $a * b^{-1} \in S$.

14. (a) Define isomorphic graphs. Prove that the following graphs are isomorphic



Or

(b) In a simple digraph, $G = (V, E)$, prove that every node of the digraph lies in exactly one strong component.

15. (a) With usual notations, prove that

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.

Or

(b) Given the following table for a sample space a random variable *X*, and a probability distribution P, find the expected value for *X*.

x_i	x_1	x_2	x_3	x_4
$X(x_i)$	5	2	3	7
$P(x_i)$	2/8	3/8	2/8	1/8

Section C (3 × 10 = 30)

Answer any **three** questions.

16. (a) Find a disjunctive normal form of

$\neg(P \vee Q) \leftrightarrow (P \wedge Q)$.

(b) Obtain the principal disfunctive normal form of $P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$.

17. Let $X = \{1, 2, \cdots 7\}$ and $R = \{(x, y) / x - y$ is divisible by 3$\}$. Show that R is an equivalence relation. Draw the graph of R.

18. Define a cyclic group. Show that every cyclic group of order n is isomorphic to the group $(Z_n, +_n)$.

19. Let A be the adjacency matrix of a digraph. Prove that the element in the ith row and j^{th} column of A^n is equal to the number of paths of length n from the ith node to the jth node.

20. State and prove Baye's theorem.

RW–966

Sub. Code
541105/545105

M.C.A. (R)/WE DEGREE EXAMINATION, NOVEMBER 2017

First Semester

COMPUTER NETWORKS

(Common for M.C.A. (R), M.C.A. (WE))

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by Network topology?

2. What is a protocol?

3. What is Framing?

4. What are the design issues of Data Link Layer?

5. What is circuit switching?

6. What is meant by Flooding?

7. What are the functions of TCP?

8. What is SNMP?

9. What is Encryption?

10. What is Substitution?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the various categories of Network.

Or

(b) Explain any two transmission media used in computer network.

12. (a) Explain the error correction and error detection with an example.

Or

(b) Explain the CSMA/CD.

13. (a) Explain virtual circuit and datagram subnets.

Or

(b) Explain the various QOS parameters.

14. (a) Explain UDP.

Or

(b) Explain the protocols used in Email.

15. (a) Explain the ECC system.

Or

(b) Explain any one symmetric key cryptography algorithm.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the functions of various layers in OSI reference model with a diagram.

17. Explain the sliding window protocol.

18. Explain Multicast routing algorithm

19. How connections are established and released by Transport protocols? Explain.

20. Explain the RSA algorithm.

RW–967

Sub. Code
541301/ 545301

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

COMPUTER GRAPHICS

(Common for M.C.A (R) / M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. Define CAD.
2. Give short notes on Image Processing.
3. Write a scaling in 2D transformation.
4. Write about General Pivot – Point Rotation.
5. Write a short notes about Polygon tables.
6. Define B-spline curves.
7. Write about translation in 3D transformation.
8. Write about viewing pipeline in 3D.
9. Write about Back-face Detection method.
10. Write a short note on the general computer animation functions.

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

All questions carry equal marks.

11. (a) Discuss briefly the Input devices.

Or

(b) Discuss about Boundary – Fill Algorithm.

12. (a) Explain about 2D – Geographical basic transformation.

Or

(b) Discuss about Cohen-Sutherland line clipping.

13. (a) Give a brief note on Quadric surfaces.

Or

(b) Discuss about spline specifications.

14. (a) Explain briefly about the scalling in 3D-Geometric transformation.

Or

(b) Discuss about transformation from world to viewing coordinates.

15. (a) Discuss about Depth-buffer method.

Or

(b) Give a brief note on Animation sequence in designing.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Explain briefly about line drawing Algorithm.

17. Discuss briefly about the composite transformation in 2D transformation.

18. Discuss about Bezier curves and surfaces.

19. Explain briefly about clipping.

20. Explain in detail about Scanline and depth sorting method.

RW–968

Sub. Code
541302/ 545302

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Computer Applications

INTERNET AND JAVA PROGRAMMING

(Common for M.C.A. (R)/M.C.A. (WE)

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What is Internet? Specify it uses.
2. What is WWW?
3. Write a short note on oops?
4. Mention some the advantages of Java Language?
5. Define the class and object.
6. What is array and its types?
7. Define Thread and its uses.
8. Define Exception.
9. Specify the structure of Java language.
10. What is stream? Specify its types.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about DNS and e-mail in internet.

Or

(b) What is Internet Protocol? Explain about FTP.

12. (a) Discuss about oops concept.

Or

(b) Distinguish between C, C++ and Java.

13. (a) Explain constructors with an example.

Or

(b) What is package? Explain in detail with an example.

14. (a) Discuss about thread and multithreaded programming in Java.

Or

(b) Explain the term Debugging and write the syntax of multiple catch statements with an example.

15. (a) Explain in detail about Java class library.

Or

(b) What is byte stream? Mention its features in java.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. (a) Explain the concept of messaging and online chatting. (5)

(b) Discuss about Telnet. (5)

17. Explain branching, looping and control statements in Java with example program.

18. Distinguish between multiple inheritance and multilevel inheritance with an example.

19. Briefly discuss about multithreaded programming with suitable example program.

20. Discuss briefly about file and its types.

RW–969

Sub. Code
541303/ 545303

M.C.A. (R/WE) DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Computer Applications

COMPUTER NETWORKS

(Common for M.C.A. (R). M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define network.

2. Name the factors that affect the performance of the network.

3. What are the responsibilities of DLL?

4. List the error detection methods.

5. Define circuit switching.

6. Define router.

7. What are the transport service primitives?

8. Define connection oriented service.

9. What do you mean by Encryption?

10. What is cryptanalysis?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) List down the uses of computer networks.

Or

(b) Explain: Internetworks.

12. (a) Discuss: DLL design issues.

Or

(b) Explain: Sliding window protocols.

13. (a) Discuss the issues involved in the implementation of connectionless service.

Or

(b) Compare virtual circuits and datagram networks.

14. (a) Discuss: Domain Name system.

Or

(b) Discuss: Remote procedure call.

15. (a) Write short notes on substitution ciphers.

Or

(b) Give RSA algorithm with example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With a neat diagram, explain the different OSI layers.

17. Give a note on CSMA protocols.

18. Discuss the congestion control algorithms.

19. Explain: WWW.

20. Explain in detail the data encryption standard algorithm.

RW–970

Sub. Code
541551/ 545551

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Applications

OBJECT ORIENTED ANALYSIS AND DESIGN

(CBCS – 2012 onwards)

[Common for M.C.A.(R)/M.C.A. (WE)]

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Give any two reasons for using object orientation.
2. What is association?
3. What are the phases of OMT?
4. What is the use of UML class diagram?
5. What is 80-20 rule?
6. What criteria would you use to eliminate a class?
7. What is cohesion?
8. What are the basic types of attributes?
9. What is the importance of developing a test case?
10. What are the different kinds of errors?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the advantages of object-oriented development?

Or

(b) What is the role of classes of objects in analysis and design?

12. (a) Describe the processes and components of the unified approach.

Or

(b) Why modeling is essential? Explain.

13. (a) Why documentation is consider as an important part of analysis? Explain.

Or

(b) Write the guidelines which helps in selecting candidate classes from the relevant and fuzzy categories of classes in the problem domain.

14. (a) Describe the activities involved in the object-oriented design process.

Or

(b) How can you refine the attributes? Discuss.

15. (a) Write a short note top-down testing.

Or

(b) Write basic activities are performed in maintenance. Explain.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain briefly about software development process.

17. Describe the booch methodology in detail.

18. Describe the different kinds of relationships among objects.

19. How can you design view layer classes? Explain.

20. Explain the impact of object orientation on testing.

RW–971

Sub. Code
541560/ 545560

M.C.A. (R/WE) DEGREE EXAMINATION, NOVEMBER 2017.

Third Semester

Computer Applications

RESOURCE MANAGEMENT TECHNIQUES

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/M.C.A. (WE))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the general mathematical formulation of Linear Programming Problem.

2. What are unrestricted variable and artificial variable?

3. What is meant by an optimality test in a transportation problem?

4. Distinguish between transportation model and assignment model.

5. Define all IPP.

6. What are the methods for solving integer programming problem?

7. Define free float.

8. What is dummy activity and when is it needed?

9. In a railway marshalling yard, goods train arrives at a rate of 30 trains per day. Assuming the inter arrival time follows an exponential distribution and the service time distribution is also exponential with an average of 36 minutes. Calculate the mean queue length.

10. What are the limitations of inventories?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Solve the following linear programming problems by using Graphical method

Maximize $z = 5x_1 + 7x_2$

Subject to

$$4x_1 + 7x_2 \leq 240$$
$$4x_1 + 6x_2 \leq 300$$
$$2x_1 + 3x_2 \geq 0$$
$$x_1, x_2 \geq 0.$$

Or

(b) Use simplex method to solve the following LPP

Maximize $z = 4x_1 + 10x_2$

Subject to

$$2x_1 + x_2 \leq 50$$
$$2x_1 + 5x_2 \leq 100$$
$$2x_1 + 3x_2 \leq 90$$
$$x_1, x_2 \geq 0.$$

12. (a) Determine the basic feasible solution to the following transportation problem using North - West Corner Rule :

	Sink					
Origin	A	B	C	D	E	Supply
P	2	11	10	3	7	4
Q	1	4	7	2	1	8
R	3	9	4	8	12	9
Demand	3	3	4	5	6	

Or

(b) Solve the assignment problem for maximizes for the profit matrix

	P	Q	R	S
A	51	53	54	50
B	47	50	48	50
C	49	50	60	61
D	63	64	60	60

13. (a) Write down the steps for Branch and Bound Algorithm to solve Integer Programming Problem.

Or

(b) Solve the following Mixed Integer Programming Problem

Maximize $Z = x_1 + x_2$

subject to

$$2x_1 + 5x_2 \leq 16$$
$$6x_1 + 5x_2 \leq 30$$
$x_2 \geq 0$, x_1 is a non-negative Integer.

14. (a) Draw the network and determine the critical path for the given data :

Jobs :	1-2	1-3	2-4	3-4	3-5	4-5	4-6	5-6
Duration :	6	5	10	3	4	6	2	9

Find the total float and free float of each activity. (3 + 2)

Or

(b) (i) Explain the three time estimates used in PERT. (3)

(ii) Distinguish between PERT and CPM. (2)

15. (a) A particular item has a demand of 9,000 units / year. The cost of one procurement is Rs. 100 and the holding cost per unit is Rs. 2.40 per year. The replacement is instantaneous and no shortages allowed. Determine

(i) the economic lot size (3)

(ii) the number of orders per year (2)

Or

(b) Customers arrive at one-man barber shop according to a Poisson process with a mean interval time of 12 min. Customers spend an average of 10 minutes in the barber's chair.

(i) What is the expected number of customers in the barber shop and in the queue? (3)

(ii) Calculate the percentage of time an arrival can walk straight into the barber's chair without having to wait. (2)

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Use Big M method to solve

Max $z = 5x_1 + 8x_2$

Subject to the constraints

$3x_1 + 2x_2 \geq 3$

$x_1 + 4x_2 \geq 4$

$x_1 + x_2 \leq 5$

$x_1, x_2 \geq 0.$

17. Solve the following transportation problem to find the optimum solution.

	To			
To / From	A	B	C	Supply
F_1	7	3	4	2
F_2	2	1	3	3
F_3	3	4	6	3
Demand	4	1	3	8

18. Solve the following Integer Mixed Integer Programming Problem by Gomory's cutting Plane Method

Maximize $z = x_1 + x_2$

Subject To

$3x_1 + 2x_2 \leq 5$

$x_2 \leq 2$

$x_1, x_2 \geq 0$ and x_1 is an integers

19. Listed in the table are the activities and sequencing requirements necessary for the completion of research report.

Activity :	A	B	C	D	E	F	G	H
Duration (Weeks) :	4	2	1	12	14	2	3	2
Immediate								
Predecessors:	E	A	B	K	-	B	F	F
Activity :	I	J	K	L	M			
Duration (Weeks) :	4	3	4	2	2			
Immediate :								
Predecessors:	F	l, L	C, G, H	D	I, L			

(a) Find the critical path (5)

(b) Find the total float and the free float for each non-critical activity. (5)

20. Arrivals at a telephone booth are considered to be poison with an average time 12 minutes between one arrival and the next. The length of telephone call is assumed to be distributed exponentially with mean 4 minutes. Calculate the following :

(a) Find the average number of persons waiting in the system. (2)

(b) What is the probability that there are 4 or more customers in the system? (2)

(c) What is the probability that an arrival will have to wait more than 10 minutes altogether to wait for phone and complete the call? (4)

(d) What is the probability of system being idle? (2)

RW–972

Sub. Code
541501/ 545501

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Applications

MULTIMEDIA AND ITS APPLICATIONS

[Common for M.C.A. (R)/M.C.A. (WE)]

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define multimedia.

2. What is meant by 24-bit color image?

3. List out the dissimilarities between PAL video and SECAM video.

4. Expand the term (a) DM (b) DPCM.

5. Why data compression is essential in multimedia?

6. What is distortion measure?

7. Write down the two frames used in video compression.

8. Define the term scalability in MPEG-2.

9. What is the use of multiplexing?

10. List the five types of AAL protocols.

Section B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the relative sizes of the LAB gamut and CMYK gamut?

Or

(b) Write short notes on color Look Up Tables. (LUTs).

12. (a) List down the advantages and disadvantages of using the interlaced video.

Or

(b) Discuss briefly about pulse code modulation.

13. (a) Describe in detail about LZW algorithm.

Or

(b) Explain in detail about vector quantization.

14. (a) What are the two types of image frames in H.261 frame sequence? Discuss.

Or

(b) Illustrate the six layers of an MPEG-1 video.

15. (a) Discuss briefly about the types of networks.

Or

(b) Write short notes on IP-multicast.

Section C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe in detail about types of file formats.

17. What is quantization? Discuss about linear and nonlinear quantization.

18. Elaborate the procedure for adaptive Huffman coding algorithm and its advantages.

19. Discuss the advantage of using as algebraic codebook in CELP coding.

20. Explain in detail about the functionalities of OSI reference model.

RW–973

Sub. Code
541502/545502

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Applications

DATA MINING AND WAREHOUSING

(Common for M.C.A. (R)/M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What do you mean by data mart in data warehouse?

2. Why we need dimension table?

3. Define data mining.

4. What is KDD?

5. Define an association rule.

6. What is a decision tree?

7. What are the different clustering techniques?

8. Write a note on neural networks.

9. List the different types of web mining.

10. What is the use of matlab tool?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) What are the various data sources for the data warehouse?

Or

(b) List out major functions and services in the data storage area.

12. (a) Give a brief account of data mining techniques.

Or

(b) Discuss about measure of similarity and dissimilarity.

13. (a) Discuss the concepts of frequent sets, confidence and support.

Or

(b) Write a algorithm for back propagation method in classification.

14. (a) What are the different phases of Birch?

Or

(b) Explain about supervised and unsupervised learning.

15. (a) How is text mining related to web mining? What are the techniques of text mining?

Or

(b) How do you handle spatial and non-spatial data, while carrying out any mining task? Explain.

Part C

$(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss about different types of schemas used in data warehouse with diagrams.

17. Explain the types of data can be mined in data mining technique.

18. Describe the working principle of the Pincer search algorithm.

19. Explain the concept of a cluster that are used in rock.

20. Write short notes on the following data mining tools

    (a) Weka

    (b) Rapid miner.

RW–974

Sub. Code
541503/ 545503

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Applications

MOBILE COMMUNICATIONS

(Common for M.C.A. (R)/M.C.A. (WE))

(CBCS – 2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define FDMA.
2. Write the pros and cons of TDMA.
3. What is Routing?
4. Where you will use GPRS?
5. Define ADHOC.
6. What is Bluetooth?
7. What is Mobile IP?
8. What is DSR?
9. What is the use of WML?
10. Define Snooping.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about frequencies for radio communication.

Or

(b) Explain about CDMA.

12. (a) Explain about GSM with example.

Or

(b) Explain about Security in GPRS.

13. (a) Explain Dynamic Host Configuration Protocol.

Or

(b) Explain about DSDV Protocol.

14. (a) Explain about Hyperlan.

Or

(b) Explain about IEEE802.11 Standard.

15. (a) Explain about Mobile TCP.

Or

(b) Explain about WAP.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about signals and its propagations.

17. How a connection is established in telecommunication system?

18. Explain Wireless LAN architecture with neat sketch.

19. Explain the network layer operations in mobile communication.

20. How a transmission is controlled and ensured in mobile communication? Discuss in detail.

RW–975

Sub. Code
541553/ 545553

M.C.A. (R)/WE DEGREE EXAMINATION, NOVEMBER 2017.

Fifth Semester

Computer Application

SOFTWARE TESTING METHODOLOGIES

(CBCS 2012 onwards)

(Common for M.C.A. (R)/M.C.A. (WE))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Write the purpose of software testing.
2. Compare verification and validation.
3. Give an example for Data Flow.
4. Why Domain Testing is required?
5. Define Regular Expression.
6. What is use of Decision Table?
7. Give any two tips for testability.
8. How will you use state graphics in Testing?
9. Define JMeter.
10. When will you use Winrunner?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about path testing.

Or

(b) Explain about taxonomy of bugs.

12. (a) Explain about Transaction Flow Testing with example.

Or

(b) Explain about Data Flow Testing.

13. (a) Explain path expression.

Or

(b) Explain the logic testing.

14. (a) Explain state graphs.

Or

(b) Explain about Graph Matrices.

15. (a) Explain about Regression Testing.

Or

(b) Explain about Testing Tools.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain about Software testing fundamentals.

17. What are the areas where domain testing is very essential?

18. How a Regular Expression is used in Software Testing?

19. Explain the applications of Graph Matrices.

20. How a website is tested? Explain with examples

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RW–976

Sub. Code
541562/ 545562

M.C.A. (R)/(WE) DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Applications

GRID COMPUTING

(CBCS – 2012 onwards)

(Common for M.C.A. (R)/M.C.A. (WE))

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is grid computing?

2. Define web services.

3. State the use of collaborative engineering.

4. List the major phases in computing and communications.

5. Expand the following (a) SAN (b) MPLS.

6. What is SOAP?

7. Why we need OGSI?

8. Define the importance of UDDI.

9. What is data caching?

10. What are two services in queuing?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) List the applications of grid computing with respect to financial benefits.

Or

(b) Explain the grid computing standards.

12. (a) Discuss suppliers and vendors of grid computing.

Or

(b) Explain the possible economic value of grid computing.

13. (a) Discuss the involvement of networks in grid computing system.

Or

(b) Summarize WSRF specifications in detail.

14. (a) Describe the grid service in detail.

Or

(b) Explain the functionalities of GSH.

15. (a) Discuss the functionality requirements of grid.

Or

(b) Illustrate the key features of OGSI.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Discuss the comparison and other approaches in grid computing.

17. Discuss briefly about the challenges faced in grid computing.

18. Describe the major components of a grid computing system from a service view.

19. Explain the following

    (a) Grid program execution services. (5)

    (b) Grid data services. (5)

20. With neat sketch explain the variety of VO service structures.